

ANALYTIC GREEN FUNCTION AND BOUND STATES FOR THE SCREENED COULOMB POTENTIAL

$$V(r) = F/r^2 + G/r + H/(r + z^2).$$

M. ZNOJIL

Institute of Nuclear Physics, Czechoslovak Academy of Sciences, 250 68 Řež near Prague, Czechoslovakia

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A complete and exact solution of the Schrödinger equation with a modified Coulomb potential is constructed by means of analytic continued fractions.

The phenomenological force $V(r) = F/r^2 + G/r + H/(r + z^2)$ is similar to the Morse potential [1] — with $G/H = -x^2 > -1$ and $F = 0$, it exhibits the repulsive barrier or attractive minimum near the coulombic attractive or repulsive centre, respectively, with the extremum $V(r_0) = H(1 - x)^2/z^2$ lying at $r_0 = z^2x/(1 - x)$. The additional centrifugal $F \neq 0$ term may be introduced to modify further the force near the origin in a way resembling, e.g., the Kratzer molecular or nuclear hard-core potentials [1]. Hence, our $V(r)$ may be of interest in atomic and molecular physics. The non-singular term $H/(r + z^2)$ simulates the field of a displaced charge so that $V(r)$ represents in fact just a simple model of the Coulomb screening: $V(r) \sim G/r$ for $r \ll 1$ while $V(r) \sim (G + H)/r$ for $r \gg 1$ and $F = 0$.

The corresponding Schrödinger equation for bound states reads

$$[-d^2/dr^2 + l(l+1)/r^2 + V(r)] \psi(r) = E\psi(r), \quad V(r) = F/r^2 + G/r + H/(r + z^2), \quad F > -(l+1/2)^2, \quad (1)$$

with $E < 0$, $\hbar = 2m = 1$, $l = 0, 1, \dots$ and $\psi \in L_2(0, \infty)$. The present letter is inspired by recent studies [2–4] of a similar problem

$$[-d^2/dx^2 + \mathcal{L}(\mathcal{L}+1)/x^2 + W(x)] \varphi(x) = \epsilon\varphi(x), \quad W(x) = h/x^2 + x^2 + \lambda x^2/(1 + gx^2), \quad (2)$$

with $g > 0$, $\mathcal{L} = 0, 1, \dots$, and $h > -(\mathcal{L} + 1/2)^2$ (non-polynomial modification of the harmonic oscillator) and by its formal equivalence to eq. (1):

Theorem 1. The change of variables

$$x = r^{1/2}/\rho, \quad \rho = (-4E)^{-1/4}, \quad \varphi(x) = r^{-1/4}\psi(r) \quad (3)$$

is an equivalence transformation between the Schrödinger eigenvalue problems (1) and (2), provided that we denote

$$\epsilon = -4G\rho^2, \quad \lambda = 4H\rho^4/z^2, \quad g = \rho^2/z^2, \quad h = 4F + (2l+1)^2 - (\mathcal{L} + 1/2)^2. \quad (4)$$

Proof. (i) Differential equations (1) and (2) coincide indeed after the change of variables (3), (4). (ii) Both of them may be interpreted as the radial Schrödinger equations with an appropriate integer l or $\mathcal{L}$ when we treat h as a function of F and vice versa. (iii) In the origin ($r \rightarrow 0$, $x \rightarrow 0$), the potentials $V(r)$ and $W(x)$ are either both regular or both singular since $l(l+1) + F > -1/4$ if and only if $\mathcal{L}(\mathcal{L}+1) + h > -1/4$. (iv) The regular solutions $\psi(r) \sim r^{1/2+[(l+1/2)^2+F]^{1/2}}$, $r \sim 0$; $\varphi(x) \sim x^{1/2+[(\mathcal{L}+1/2)^2+h]^{1/2}}$, $x \sim 0$ are properly behaved both in the origin and, respectively, for $0 < r < \infty$ and $0 < x < \infty$. (v) In the asymptotic re-

gion ($r \rightarrow \infty, x \rightarrow \infty$), we may postulate the standard boundary conditions (bounded character of the wave function) in both cases. In accord with the elementary oscillation theorems [1] it defines the complete solution of (1) or (2) since

$$\psi(r) \sim Ae^{-kr} + Be^{+kr}, \quad r \gg 1, \quad k^2 = -E; \quad \varphi(x) \sim Ce^{-x^2/2} + De^{+x^2/2}, \quad x \gg 1,$$

respectively. This completes the proof since there is a one-to-one correspondence between the solutions. Eq. (4) or its inverse

$$F = h/4 + (\mathcal{L}/2 + 1/4)^2 - (l + 1/2)^2, \quad E = -1/4g^2, \quad G = -\epsilon/4g, \quad H = \lambda/4g^2, \quad r = gx^2, \quad \psi(r) = g^{1/4}x^{1/2}\rho(x) \quad (5)$$

(with $z^2 = 1$ for simplicity) are one-to-one mappings.

Let us describe now some of the less obvious consequences of theorem 1 in more detail.

1. Tables. By an extrapolation or interpolation of the results of, say, ref. [2], we may generate the numerical tables of the binding energies $E = E(G, H)$ pertaining to eq. (1). A sample is shown in tables 1 and 2 where $l = 0$ and $F = -3/16$. In the same spirit, the various asymptotic or perturbative expansions and techniques of ref. [2] may be re-interpreted as well.

2. Algorithms. In ref. [3], the multiplication of eq. (2) by $1 + gx^2$ from the left was used to simplify its representation in the harmonic-oscillator basis $|n\rangle, n = 0, 1, \dots$. This basis may be used also with a non-interer $\mathcal{L}$ — we put $h = 0$ and get $\mathcal{L} = \mathcal{L}(l, F)$ from (4) or (5). The definition

$$(-d^2/dx^2 + \mathcal{L}(\mathcal{L} + 1)/x^2 + x^2)\langle x|n\rangle = \epsilon_n\langle x|n\rangle, \quad \epsilon_n = 4n + 2\mathcal{L} + 3, \quad n = 0, 1, \dots,$$

implies that

$$x^2|n\rangle = b_{n-1}|n-1\rangle + \frac{1}{2}\epsilon_n|n\rangle + b_n|n+1\rangle, \quad b_n = [(n+1)(n+\mathcal{L}+3/2)]^{1/2},$$

Table 1

Coulombic strengths G giving the prescribed energy of the ground state.

H	G		
	$E = -0.25$	-0.01	-0.0025
50	-3.3671871		
0.05	-0.3618325	-0.2379236	-0.2320291
0.005		-0.0524404	-0.0279575

Table 2

Coulombic strengths G giving the prescribed energy of the m th excited state, $H = 0.005$.

m	G		
	$E = -0.01$	-0.0025	-0.0001
1	-0.1540186	-0.0794035	-0.01984185
2	-0.2539023	-0.1291448	-0.02953019
3	-0.3542460	-0.1795182	-0.03985890

and gives eq. (2) the form

$$\sum_{n=m-1}^{m+1} \langle m|(1+gx^2)|n\rangle (G+H+\epsilon_n/4g) \langle n|\varphi\rangle = H\langle m|\varphi\rangle, \quad m=0, 1, \dots, \quad (6)$$

where we have preserved the energy variable $g = 1/(-4E)^{1/2}$ and replaced ϵ and λ by the original couplings of eq. (1), namely, by G and H .

Eq. (6) may systematically be truncated and solved numerically, as proved in ref. [4]. When re-written in the form of recurrences

$$Q_{m-1} z_{m-1} + Q_{mm} z_m + Q_{m+1} z_{m+1} = 0, \quad m=0, 1, \dots, \quad Q_{-1} = 0,$$

$$Q_{mm} = 1 + g\epsilon_m/2 - H/(G+H+\epsilon_m/4g), \quad Q_{m+1} = Q_{m+1m} = gb_m, \quad z_m = \langle m|\varphi\rangle [\epsilon_m + 4g(G+H)], \quad (7)$$

it even leads to the compact formula

$$z_n = \frac{z_0}{(-g)^n b_0 b_1 \dots b_{n-1}} \det \begin{pmatrix} Q_{00} & Q_{01} & 0 & \dots & 0 \\ \vdots & & & & \vdots \\ 0 & \dots & 0 & Q_{n-2, n-1} & Q_{n-1, n-1} \end{pmatrix}, \quad (8)$$

which defines the projections of φ in an explicit way, unless the binding energy $g = g_0$ satisfies the condition

$$G+H = -\epsilon_M/4g_0, \quad M \geq 0. \quad (9)$$

The eigenstate $\psi(r)$ is easily expressible in terms of $\varphi(x) = \sum_n \langle x|n\rangle \langle n|\varphi\rangle$ by means of eq. (5).

3. The particular polynomial solution. Let us assume now that eq. (9) is satisfied. In such an exceptional case, the projection $\langle M|\varphi\rangle$ is defined simply by the M th row of eq. (6) so that $z_M = 0$. We may also put $Q_{MM} z_M = -H\langle M|\varphi\rangle$ in eq. (7). As a consequence, eqs. (6) or (7) become decoupled into the two independent subsystems $m=0, 1, \dots, M-1$ and $m=M+1, M+2, \dots$. Structure of the second one coincides with eq. (6) or (7) (with $m \geq M+1$ only) and need not be considered separately. The first set is more interesting since it is finite: It possesses a nontrivial solution $\langle 0|\varphi\rangle, \langle 1|\varphi\rangle, \dots, \langle M-1|\varphi\rangle$ if and only if

$$\det \begin{pmatrix} Q_{00} & Q_{01} & 0 & \dots & 0 \\ \vdots & & & & \vdots \\ 0 & \dots & 0 & Q_{M-2, M-1} & Q_{M-1, M-1} \end{pmatrix} = 0, \quad M \geq 1. \quad (10)$$

In analogy with ref. [3], we have

Theorem 2. For each fixed $g_0 > 0$, the algebraic system of eqs. (9) and (10) has M real (possibly degenerate) solutions G, H .

Proof. Eq. (9) may be inserted into the first M rows of (6). After a slight modification, this becomes an eigenvalue problem

$$\begin{pmatrix} \alpha_0 & \beta_0 & & & \\ \beta_0 & \alpha_1 & \beta_1 & & \\ & \ddots & \ddots & & \\ & & \ddots & \ddots & \\ & & & \beta_{M-2} & \alpha_{M-1} \end{pmatrix} \begin{pmatrix} J_0 \\ J_1 \\ \vdots \\ J_{M-1} \end{pmatrix} = -H \begin{pmatrix} J_0 \\ J_1 \\ \vdots \\ J_{M-1} \end{pmatrix}, \quad (11)$$

$$J_m = z_m/(M-m)^{1/2}, \quad \alpha_m = (M-m)(1/g + \epsilon_m/2), \quad m=0, 1, \dots, M-1,$$

$$\beta_m = b_m / [(M - m)(M - m - 1)]^{1/2}, \quad m = 0, 1, \dots, M - 2,$$

where the left-hand side matrix is H -independent, real and symmetric.

As a consequence, the set of eqs. (8)–(10) defines an elementary solution to the Schrödinger eq. (1) or (2). The different roots of eq. (10) define the different potentials with the fixed “asymptotic charge” $G + H$ [cf. eq. (9)]. For $\mathcal{L} = h = 0$ at least, the value of H is either zero (= pure Coulomb) or negative [3]. The G may have both signs.

4. The general analytic solution. Having in mind eq. (8), it remains for us to specify the energy eigenvalues $g = g_0$ in the general potential. In connection with eq. (2), this problem was thoroughly discussed in ref. [4]. Here, we may therefore just quote the results after their easy re-interpretation.

We shall use an auxiliary set of the continued fractions $f_k(g)$ defined for almost all $g > 0$ by the recurrences

$$f_k(g) = 1/[Q_{kk} - g^2 b_k^2 f_{k+1}(g)], \quad k = 0, 1, \dots, \quad (12)$$

from the initialization $f_{k_{\max}} = 0$ in the limit $k_{\max} \rightarrow \infty$.

Theorem 3. Unless eq. (9) is satisfied, the continued fraction $f_{n_0}(g)$ with $n_0 = 0$ is convergent and its poles $g = g_0$ define the binding energies $E = E_0 = -1/4g_0^2$ of the Schrödinger eq. (1),

$$1/f_{n_0}(g_0) = 0.$$

Similarly, the normalizable wave function is given by the convergent series

$$\psi(r) = r^{1/4} \sum_{n=n_0}^{\infty} \frac{\langle x(r)|n \rangle z_n}{\epsilon_n + 4g_0(G + H)}, \quad x(r) = (r/g_0)^{1/2}, \quad z_n = (-g_0)^{n-n_0} z_{n_0} \prod_{i=n_0}^{n-1} b_i f_{i+1}(g_0),$$

$$n = n_0 + 1, n_0 + 2, \dots \quad (13)$$

If eq. (9) holds, then either the solution (13) terminates ($z_M = z_{M+1} = \dots = 0$, cf. the preceding paragraph) or we have to choose $n_0 = M$.

Proof. The proof employs the method of ref. [5] and is the same as in ref. [4].

The choice of n_0 is a technical complication connected with the necessity to regularize eq. (12). It may easily be avoided in:

Theorem 4. The exact binding energies $E_0 = -1/4g_0^2$ and the first N coefficients z_n in eq. (13) may be calculated from a finite set of the linear algebraic equations

$$\begin{pmatrix} Q_{00} & Q_{01} & 0 & & & & & 0 \\ Q_{10} & Q_{11} & Q_{12} & 0 & & & & 0 \\ \cdot & & & & & & & \cdot \\ \cdot & & & & & & & \cdot \\ \cdot & & & & & & & \cdot \\ 0 & & \dots & 0 & Q_{N-1 \ N-2} & Q_{N-1 \ N-1} & Q_{N-1 \ N} & z_{N-1} \\ 0 & & \dots & 0 & Q_{N \ N-1} & 1/f_N(g) & & z_N \end{pmatrix} \begin{pmatrix} z_0 \\ z_1 \\ \cdot \\ \cdot \\ \cdot \\ z_{N-1} \\ z_N \end{pmatrix} = 0, \quad N \geq 0. \quad (14)$$

Proof follows from theorem 3 and from eqs. (7) and (8).

5. Approximations and both-sided bounds for the measurable quantities. To accelerate the convergence of the continued fractions, we may use their asymptotic expansions [4] and obtain our most important result:

Theorem 5. To obtain the approximate solution to eq. (14), we may replace $f_N(g)$ by its approximant

$$f_N^{(c)}(g) = [1 - (gN)^{-1/2} + cN^{-1} + O(N^{-3/2})]/b_{N-1}, \quad N \gg 1, \quad (15)$$

with any $c = O(1)$. With the pair $c = c_1$ and $c = c_2$ such that $1 \ll c_1 \ll 2(N/g)^{1/2}$ and $c_2 \gg 2(N/g)^{1/2}$ or $c_2 \ll -1$, we obtain the pairs of approximants (to the energy and expansion coefficients z_n) which converge to the exact values from both sides in the limit $N \rightarrow \infty$.

Proof is given in ref. [4]. In brief, it follows from the $k \gg 1$ asymptotic form of eq. (12), $f_k(g) = 1/[2gk - g^2k^2f_{k+1}(g) + O(1)]$. This mapping $f_{k+1} \rightarrow f_k$ is only weakly k -dependent and has a unique point of accumulation given by eq. (15) with some particular $c = O(1)$. Hence, any $O(1/N)$ change in the exact initialization $f_{N+1}^{(c_0)}$ of the recurrences (12), i.e., variation $\delta c = c - c_0$, induces a similar change in $f_{N-k}(g)$, $k = 0, 1, \dots$ since $\partial f_k^{(c)}(g)/\partial c \neq 0$. The modified secular equation $[0 = 1/f_0(g_0) \rightarrow 0 = 1/f_0^{(c)}(g_0 + \delta g)]$ gives $\delta g = A \delta c$ where $A \doteq -(\partial/\partial c)[1/f_0^{(c)}(g_0 + \delta g)]/[(\partial/\partial g)(1/f_0(g_0))]$ is small [$\leq O((1/N)^{1/2})$ for $c \neq 2(N/g)^{1/2} \pm O(1)$] and non-zero near g_0 .

6. Conclusions. The solution of eq. (1) and most of its properties follow immediately from the re-interpretation of the results of refs. [2–4]. In an obvious way, this (theorem 1) may also be generalized to the zero-dimensional interpretation of eq. (1) [with $r \in (-\infty, \infty)$, $l = -1$ and $0 = \text{parity}$, etc.] and/or to more general potentials [6].

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